

SARDAR PATEL UNIVERSITY**EXAMINATION APRIL 2018****BCA IInd - SEMESTER (CBCS) (REG & NC)****(June 2011 BATCH ONWARDS)****Saturday, Date: 31-03-2018****Session: Morning Time: 10:00 A.M. to 1:00 P.M.****Subject Code: US02CBCA01****Total Marks: 70****Subject Title : Advanced C Programming and Introduction to Data Structures****Q1. Multiple Choice Questions. (Attempt all) [10]**

1. Which of the following defines a pointer variable to character?
A. char &pa; C. char =pa;
B. char *pa; D. char &&pa;
2. Which of the following is not a 'C' dynamic memory allocation function?
A. realloc() C. malloc()
B. alloc() D. calloc()
3. Which of the following is not a derived data type?
A. array C. structure
B. union D. integer
4. f = fopen(filename, "r");
Referring to the code above, what is the proper definition for the variable f?
A. File *f; C. struct FILE *f;
B. FILE *f; D. int FILE *f;
5. Which of the following can be used to create a new type that can be used anywhere a type is permitted?
A. struct C. typedef
B. union D. new
6. When new data are to be inserted into a stack data structure, but there is no available space, this situation is usually called _____.
A. Underflow C. Overflow
B. Housefull D. Available
7. Which of the following is an operation of a Stack data structure?
A. Rear C. Front
B. POP D. TOP
8. A linked list in which last node pointing to the first node is known as _____.
A. Doubly Linked List C. Circular Linked List
B. Singly Linked List D. All of the above
9. Which of the following is TRUE for a Queue data structure?
A. FIFO C. Linear
B. Non-Primitive D. All of the above
10. A data structure in which insertion of an element occurs at one end and deletion of an element occurs at other end is known as _____.
A. Queue C. Tree
B. Stack D. Graph

Q2. Answer the following Short Questions (Attempt any TEN)

[20]

1. Explain how Compile time and Runtime memory allocation process differ?
2. List different pointer declaration style. Which one is preferable?
3. Differentiate between '*' and '&' operators in pointers.
4. Explain fclose() function with example.
5. List all modes in which you can open a file. Explain any one in brief.
6. Define a structure called **cricket** which contains an integer quantity called **crank** and strings called **name**, **country** with length 40 and length 60 respectively. Create another structure called **score** within structure **cricket** which contains integer quantities called **total_runs**, **total_four** and **total_six**.
7. List applications of Stack data structure.
8. Give 4 examples of Non-Primitive data structure.
9. Give representation of Stack data structure and Explain what does TOP denote?
10. List applications of Queue.
11. Explain Dequeue in brief.
12. Differentiate between Singly Linked List and Doubly Linked List.

Q3.a. Explain Pointer to an Array using appropriate examples. [05]

b. Explain any 2 functions used for Dynamic Memory Allocation. [05]

OR

Q3.a. Define Pointer variable. Explain how to declare and initialize pointer variable? Explain how can we access value of a variable through pointer type variable? [05]

b. Write brief note on pointer to pointer with example. [05]

Q4.a. Explain concept of Array of Structure suitable example. [05]

b. What is Union? Explain Union definition, declaration and assigning values to members of Union with suitable example. [05]

OR

Q4.a. Explain getw() and putw() file handling functions with example. [05]

b. Differentiate between : Structure and Union [05]

Q5.a. Explain in brief all operations that can be performed on a Data Structure. [04]

b. Write algorithms for searching and updating an element on a Stack. [06]

OR

Q5.a. Write short note on Linear Data Structures. [04]

b. Write algorithms for inserting an element on a Stack and deleting an element from Stack. [06]

Q6. What is Linked List? Write an algorithm to insert an element at the end of a Singly Linked List. [10]

OR

Q6. What is Queue? Give representation of Queue. Write an algorithm to insert an element into a Queue and delete an element from a Queue. [10]